

SC2000/CZ2100

Probability & Statistics

Week 4

Ch 6. Probability Distributions

- Random variables – Discrete
 - Continuous
- Discrete Probability distributions – Binomial, Poisson, Geometric
- Continuous Probability distributions – Uniform, Exponential, Normal

Random Variables

The outcomes of random experiments commonly takes on numerical values.

If the outcomes are not numerical, they can be represented in terms of numbers (this facilitates mathematical analysis).

A random variable (r.v.) is a variable that has a numerical value which is defined on or determined by the outcomes or events of an experiment. The numerical value cannot be predicted exactly but can be described by its probability.

Random Variables

Random variables can be discrete or continuous.

Eg: The no. of seeds germinating from a flower pot. Possible values for X are 0, 1, 2, ... (discrete).

The maximum daily temperature in Singapore. Possible values are 23 – 35°C, e.g. 26.1276°C (continuous).

The response to questions with “Yes”, “No”, or “Don't know” answers. This are not a r.v. (the values are not numerical).

Let Y be the number of “Yes”s.
 Y is a discrete r.v.

Random Variables

Expected Value of a r.v.

For a given set of observed data $\{x_1, x_2, \dots, x_n\}$, one of the central tendency measure is the arithmetic average:

$$\bar{X} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$\bar{X}$ is calculated based on a sample of n observed values.

If we know the probability of occurring associated with each value of a r.v., then we can obtain the weighted average which is known as the expectation of the r.v.

Random Variables

Expected Value of a r.v.

The probability distribution or probability mass function (pmf) of a r.v. X is given as:

Values of X	x_1	x_2	$\dots$	x_k
Probabilities	p_1	p_2	$\dots$	p_k

Its expected value is defined as:

$$E(X) = x_1 p_1 + x_2 p_2 + \dots + x_k p_k = \sum_{i=1}^k x_i p_i$$

Also denoted by μ_x

Random Variables

Eg: (Decision Analysis)

An oil exploration company has a lease for which it must decide on one of the following options:

1. Sell now and can get \$125K.
2. Hold for a year and if oil prices rise (prob = 0.6) it can sell for \$300K or if oil prices fall (prob = 0.4) it can get \$100K.
3. Drill now. The cost of drilling is \$200K and drilling will lead to one of the following outcomes:

Well type	Dry	Wet	Gusher
Probability	0.5	0.4	0.1
Profit	\$0	\$400K	\$1500K

What should the company do?

Random Variables

Solution: Let X be the financial gain.

Option 1 (sell now): financial gain = \$125K.

Option 2

(hold for 1 yr):

X	300	100
Probability	0.6	0.4

$$\therefore E(X) = 300 \times 0.6 + 100 \times 0.4 = \$220K$$

Option 3, X = profit – drilling costs (\$200K)

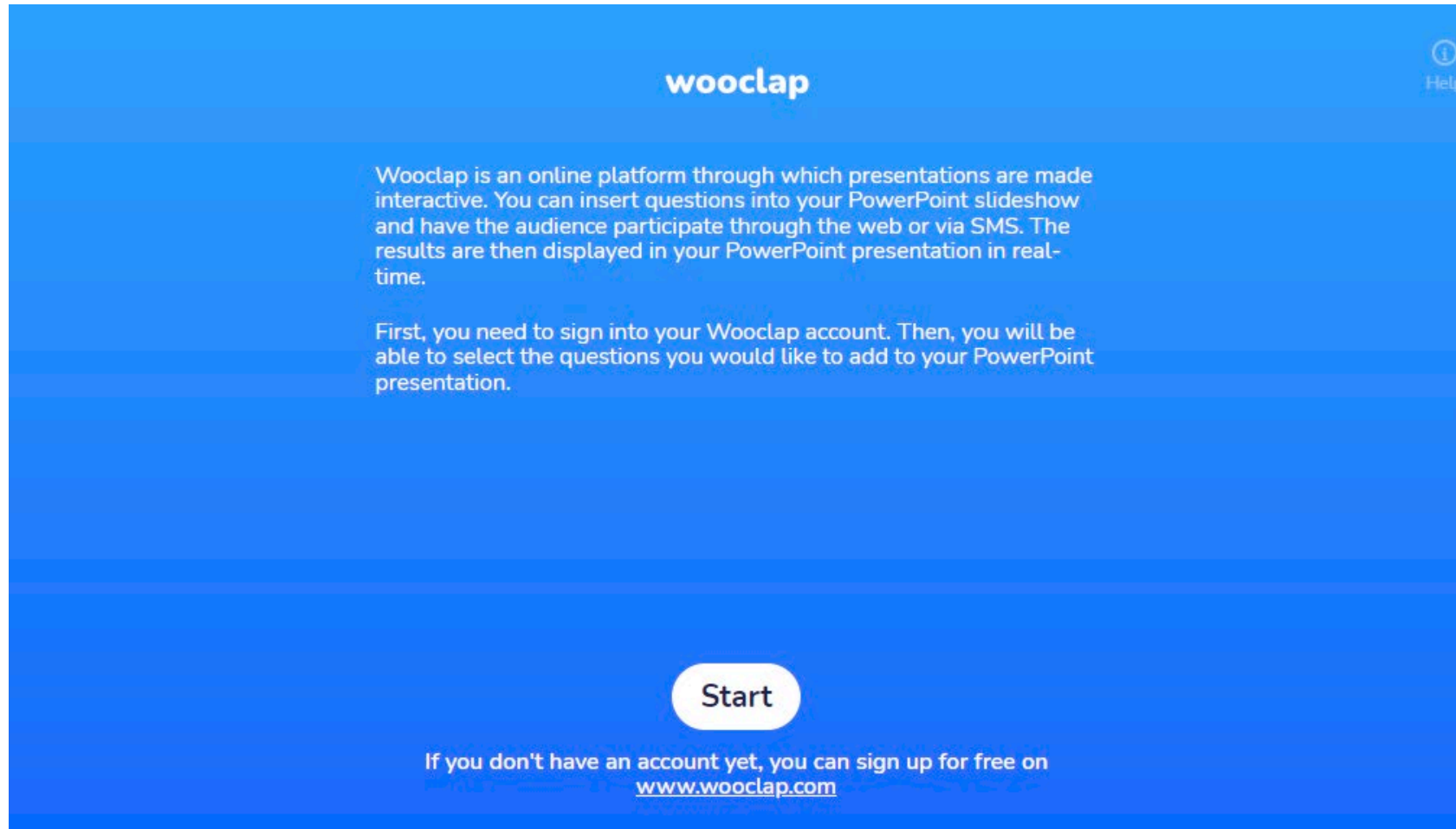
X	0 – 200K	400K – 200K	1,500K – 200K
Probability	0.5	0.4	0.1

$$\therefore E(X) = -200 \times 0.5 + 200 \times 0.4 + 1300 \times 0.1 = \$110K$$

Best decision is to hold for a year and then sell.

Given the pmf of X below, if $Y=(X-1)^2$, find $E[Y]$.

x	0	1	2
$P_X(x)$	0.2	0.5	0.3

A screenshot of a presentation slide from the Wooclap platform. The slide has a blue background with white text. At the top, the 'wooclap' logo is centered, and a 'Help' icon is in the top right corner. The main text describes Wooclap as an interactive presentation platform. Below this, it instructs users to sign into their account to add questions to their PowerPoint presentation. A large 'Start' button is centered at the bottom. At the very bottom, it provides a link for users who don't have an account yet.

wooclap Help

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Start

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Given the pmf of X below, if $Y=(X-1)^2$, find $E[Y]$.

x	0	1	2
$P_X(x)$	0.2	0.5	0.3

Random Variables

Variance of a r.v.

Recall that variance is a measure of variability. For a sample of n independent observations the sample variance is obtained using:

$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{X})^2}{n-1} \quad \text{or} \quad \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right]$$

The variance of a r.v. X is defined as

$$\begin{aligned} \text{var}(X) &= p_1(x_1 - \mu)^2 + p_2(x_2 - \mu)^2 + \cdots + p_n(x_n - \mu)^2 \\ &= \sum_{i=1}^n p_i(x_i - \mu)^2 = E[(x - \mu)^2] \end{aligned}$$

where p_i = probability that x_i occurs.

Random Variables

It represents the theoretical limit of the sample variance s^2 as the sample size n becomes very large. $\text{Var}(X)$ is often denoted by σ^2 .

An alternate formula for $\text{var}(X)$ is as follows:

$$\begin{aligned}\text{var}(X) &= \sum p_i (x_i^2 - 2x_i\mu + \mu^2) \\ &= \sum p_i x_i^2 - 2\mu \sum p_i x_i + \mu^2 \sum p_i \\ &= E[X^2] - \mu^2 \quad \text{or} \quad E[X^2] - (E[X])^2\end{aligned}$$

Random Variables

Eg

Let the random variable X be the number of male students in a group from a class of size 5. The probability distribution of X is

X	0	1	2	3	4	5
$P(X = x)$	1/32	5/32	10/32	10/32	5/32	1/32

What is the expected number of males $E(X)$ in the group, and what is the standard deviation, σ , of X ?

Random Variables

Solution $E(X) = \sum_{i=1}^n x_i P(x_i)$

$$= 0 \times \frac{1}{32} + 1 \times \frac{5}{32} + 2 \times \frac{10}{32} + 3 \times \frac{10}{32} + 4 \times \frac{5}{32} + 5 \times \frac{1}{32}$$
$$= \frac{80}{32} = 2.5 = \mu$$

i.e. on average such groups have 2.5 males.

$$\text{var}(X) = \sum x^2 P(x) - \mu^2$$
$$= \left(0^2 \times \frac{1}{32} + \dots + 5^2 \times \frac{1}{32}\right) - (2.5)^2$$
$$= 7.5 - 2.5^2 = 1.25$$

So $\sigma = \sqrt{\text{var}(X)} = \sqrt{1.25} \approx 1.12$

Given the pmf of X below, if $Y=(X-1)^2$, find $\text{Var}[Y]$.

x	0	1	2
$P_X(x)$	0.2	0.5	0.3



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Given the pmf of X below, if $Y=(X-1)^2$, find $\text{Var}[Y]$.

x	0	1	2
$P_X(x)$	0.2	0.5	0.3

Random Variables

Empirical quantity (m observed data)	Theoretical quantity (mathematical d.r.v.)	Remarks
Relative freq of x_i is $\frac{f_i}{m}$	$P[X = x_i] = p_i$	$\frac{f_i}{n} \rightarrow p_i$ as $n \rightarrow \infty$
$\sum_i \frac{f_i}{m} = 1$	$\sum_i p_i = 1$	
Mean: $\bar{X} = \sum_i \frac{f_i}{m} x_i$	Expectation: $E[X] = \mu = \sum_i p_i x_i$	$\bar{X} \rightarrow \mu$ as $n \rightarrow \infty$
Variance: $s^2 = \sum_i \frac{(x_i - \bar{X})^2 f_i}{m - 1}$	Var[X]: $\sigma^2 = \sum_i (x_i - \bar{X})^2 p_i$	$s^2 \rightarrow \sigma^2$ as $n \rightarrow \infty$

Random Variables

Expected Value and Variance for a Function of r.v.

(Linear transformation of r.v.)

If $Y = a + bX$ where X is a r.v. and a and b are known constant values, then

$$E(Y) = a + b E(X)$$

and

$$\text{var}(Y) = b^2 \text{var}(X)$$

so

$$\sigma_y = \sqrt{b^2 \text{var}(X)} = \sqrt{b^2 \sigma_x^2} = b \sigma_x$$

Random Variables

Similarly if $T = a + bX + cY$ where X and Y are uncorrelated r.v. and a , b and c are known constants, then

$$E(T) = a + bE(X) + cE(Y)$$

and

$$\text{var}(T) = b^2 \text{var}(X) + c^2 \text{var}(Y)$$

Random Variables

Eg: A company makes products for local and export markets. The number of sales for next year are estimated as follows:

local, X units	1,000	3,000	5,000	10,000
probability	0.1	0.3	0.4	0.2

export, Y units	300	500	700
probability	0.4	0.5	0.1

$$\begin{aligned}\text{Hence } E(X) &= 1000 \times 0.1 + 3000 \times 0.3 + \\ &\quad 5000 \times 0.4 + 10000 \times 0.2 \\ &= 5000 \text{ (= expected local sales)}\end{aligned}$$

Random Variables

$$\begin{aligned} E(Y) &= 300 \times 0.4 + 500 \times 0.5 + 700 \times 0.1 \\ &= 440 \quad (= \text{expected export sales}) \end{aligned}$$

Suppose the company makes a profit of \$2000 on each unit sold on the local market and \$3500 on each exported unit.

Hence the total profit is $T = 2000X + 3500Y$

And the expected profit is

$$\begin{aligned} E(T) &= 2000E(X) + 3500E(Y) \\ &= 2000 \times 5000 + 3500 \times 440 \\ &= \$11,540,000 \end{aligned}$$

$\Rightarrow$ this is next year's estimated profit.

Random Variables

Eg: A component is made by cutting a piece of metal to length X and trimming it by amount Y . Both processes are somewhat imprecise.

Nett length $T = bX + cY$ with $b = 1$ and $c = -1$.

$$E(T) = bE(X) + cE(Y) = E(X) - E(Y)$$

$$\text{Var}(T) = b^2 \text{var}(X) + c^2 \text{var}(Y)$$

↑
Note

i.e. $\text{var}(T)$ is greater than either $\text{var}(X)$ or $\text{var}(Y)$, even though $T = X - Y$, because both X and Y contribute to the variability in T .

Prob. Distribution of Discrete r.v.

Special Probability Distributions

- Binomial Distribution
- Poisson Distribution
- Geometric Distribution

Prob. Distribution of Discrete r.v.

The list of possible values that a d.r.v. X can take and their probabilities is called the discrete probability distribution (also known as probability mass function, pmf) for X .

Eg:

Let the random variable X be the number of girls in a family of 3 children.

Possible values:

$X = 3$	<i>GGG</i>		
$X = 2$	<i>GGB</i>	<i>GBG</i>	<i>BGG</i>
$X = 1$	<i>BBG</i>	<i>BGB</i>	<i>GBB</i>
$X = 0$	<i>BBB</i>		

Prob. Distribution of Discrete r.v.

Assume that the 8 outcomes are equally likely,

x_i	0	1	2	3
Probability Distribution $P(X=x_i)$	1/8	3/8	3/8	1/8
Cumulative Distribution $P(X \leq x_i)$	1/8	4/8	7/8	1

In general, we write:

$$P(X = x_i) = p_i \text{ for } i = 1, \dots, k, \text{ and } 0 \leq p_i \leq 1$$

Notation convention: we often use capital letters for random variables and small letters for specific values.

Prob. Distribution of Discrete r.v.

Eg: Let X denote the sum of the results of 2 dice thrown.

Outcomes:

1,1	2,1	3,1	...	6,1
1,2	2,2	3,2	...	6,2
⋮	⋮	⋮		⋮
1,6	2,6	3,6	...	6,6

36 outcomes and the values of X are 2, 3, ..., 12.

Assuming that the outcomes are equally likely, i.e. the probability of each outcome is $1/36$, the probability distribution of X is

X	2	3	4	...	10	11	12
$P(X = x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	...	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Exercise: find the cumulative distribution.

Prob. Distribution of Discrete r.v.

Special Probability Distributions

- Binomial Distribution
- Poisson Distribution
- Geometric Distribution

Prob. Distribution of Discrete r.v.

Binomial Distribution

- Consider n Bernoulli trials, where each trial is an “experiment” with exactly 2 possible outcomes, “success” and “failure”.
- Assume that the probability of success (S) is the same for all trials, $P(S) = p$, and the probability of failure $P(F) = 1 - p$.
- Assume also that trials are independent, i.e., the probability for any given combination of successes and failures can be obtained by multiplying the probabilities for each trial outcome.

Prob. Distribution of Discrete r.v.

Eg: Consider 5 trials, the probability,

$$\begin{aligned}P(SSFSF) &= p \cdot p \cdot (1-p) \cdot p \cdot (1-p) \\ &= p^3(1-p)^2\end{aligned}$$

The probability of obtaining any 3 successes and 2 failures in 5 trials, i.e. $SSSFF$, $SSFSF$, ... etc., is $p^3(1-p)^2$ for each of the different ways this could occur.

Prob. Distribution of Discrete r.v.

The number of distinct "arrangements" of 3 successes and 2 failures can be calculated using the binomial coefficient $\binom{n}{x}$ or nC_x

The binomial coefficient is defined as: ${}^nC_x = \frac{n!}{x! (n-x)!}$
where $n! = n \times (n-1) \times \dots \times 2 \times 1$

In this example, $\binom{5}{3} = \frac{5!}{3!2!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{(3 \times 2 \times 1)(2 \times 1)} = 10$

so there are 10 distinct ways to obtain 3 successes in 5 trials, with each arrangement having a probability of $p^3(1-p)^2$.

Prob. Distribution of Discrete r.v.

Let X be the r.v. equal to the total number of successes in n trials. To calculate the probability of obtaining x successes, we have:

$$P(X = x) = \binom{n}{x} \cdot p^x \cdot (1-p)^{n-x}$$

arrangements of x S 's and $(n-x)$ F 's prob. of x S 's prob. of $(n-x)$ F 's

The distribution of the count of successes is called the binomial distribution with two parameters, n and p .

We say $X \sim B(n, p)$.

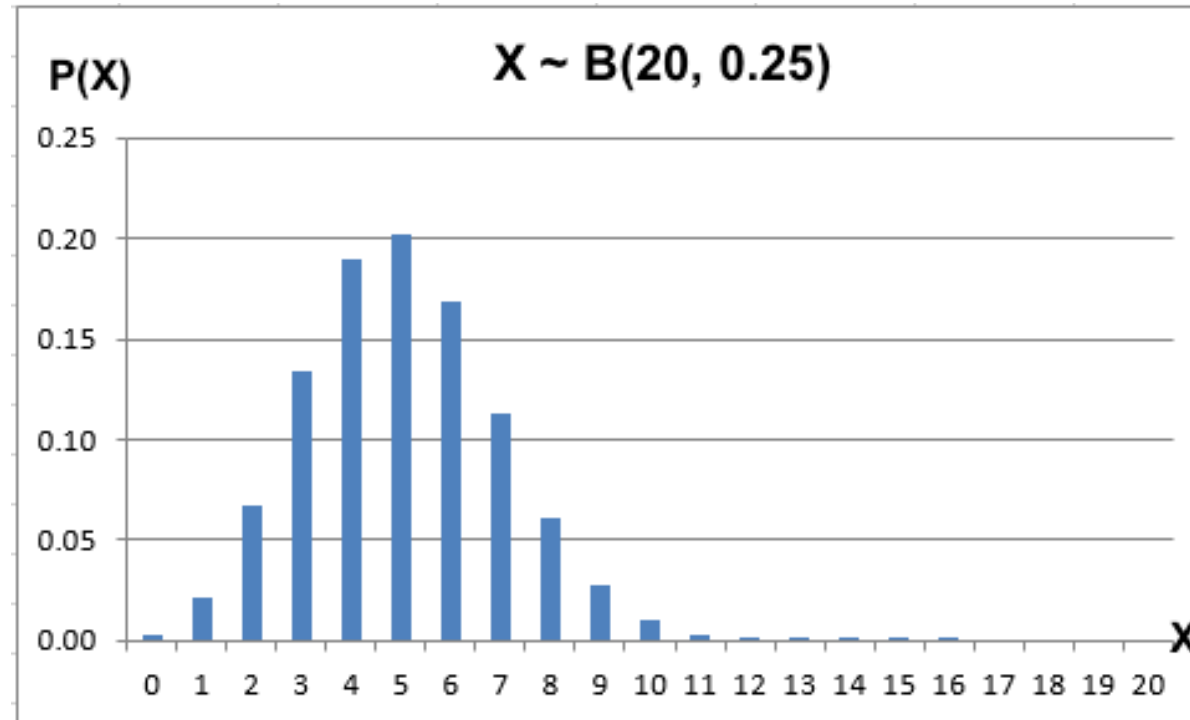
X	0	1	...	n
$P(X=x)$	${}^nC_0 p^0 (1-p)^n$	${}^nC_1 p^1 (1-p)^{n-1}$	...	${}^nC_n p^n (1-p)^{n-n}$

Prob. Distribution of Discrete r.v.

Binomial Distribution – $X \sim B(n,p)$

$$P(X = x) = \binom{n}{x} \cdot p^x \cdot (1-p)^{n-x}$$

Example:



Prob. Distribution of Discrete r.v.

Mean and variance of $X \sim B(n, p)$

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Recall:
 $E[X] = \sum x p(x)$

The mean of X is given by:

$$E[X] = \sum_{x=0}^n x \binom{n}{x} p^x (1 - p)^{n-x} = np$$

And the variance is:

$$Var[X] = E[X^2] - E[X]^2 = np(1 - p) = npq$$

where $q = (1 - p)$

Prob. Distribution of Discrete r.v.

Example:

A football team plays 3 games. Assume each game is a Bernoulli trial with $\text{prob}(\text{win}) = 0.5$. Let the r.v. X be the number of wins. What is the probability that the team will win exactly 2 games?

Solution

X has binomial distribution with $n = 3$ and $p = 0.5$, outcome Win(W) or Lose(L) on each trial. i.e.

$$X \sim B(3, 0.5)$$

Prob. Distribution of Discrete r.v.

Using the formula for Binomial probabilities:

$$\begin{aligned}P(X = x) &= \binom{n}{x} p^x (1 - p)^{n-x} \\&= \binom{3}{2} 0.5^2 (1 - 0.5)^1 \\&= 3/8\end{aligned}$$

There are 10 blue and 30 green balls in a bag. A ball is random picked, the colour is noted and it is put back into the bag. This process is repeated 10 times. Let X be the number of times a blue ball was picked, Calculate $E[X]$ and $\text{Var}[X]$.

$X \sim B(n, p)$ where $n = 10$ and $p = 10/40$

$$E[X] = np$$

$$\text{Var}[X] = npq, \text{ where } q = 1 - p$$

Prob. Distribution of Discrete r.v.

Example:

A quality control system selects a sample of 10 items from each batch of products for testing. If 2 or more of the items are defective the whole batch is rejected.

If the probability of an item being defective is 0.05, what is the probability of

- (i) having 2 defectives in the sample?
- (ii) the batch being rejected?

Prob. Distribution of Discrete r.v.

Soln: Let X be the number of defectives in the sample of $n = 10$ items. $X \sim B(10, 0.05)$

$$(i) P(X=2) = {}^{10}C_2 (0.05)^2 (0.95)^8$$

$$\begin{aligned}(ii) P(\text{reject batch}) &= P(X \geq 2) \\ &= 1 - P(X < 2) \\ &= 1 - P(X=0) - P(X=1) \\ &= 1 - 0.5987 - 0.3151 \\ &= 0.0861\end{aligned}$$

Prob. Distribution of Discrete r.v.

Special Probability Distributions

- Binomial Distribution
- Poisson Distribution
- Geometric Distribution

Prob. Distribution of Discrete r.v.

Poisson Distribution – Notation: $X \sim \text{Pois}(\mu)$

- Let r.v. X be the number of “successes” in a given time interval where X is a non-negative integer
- The r.v. X is a Poisson r.v. with probability distribution given by

$$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$$

where μ is the average number of successes and e is a constant = 2.71828...

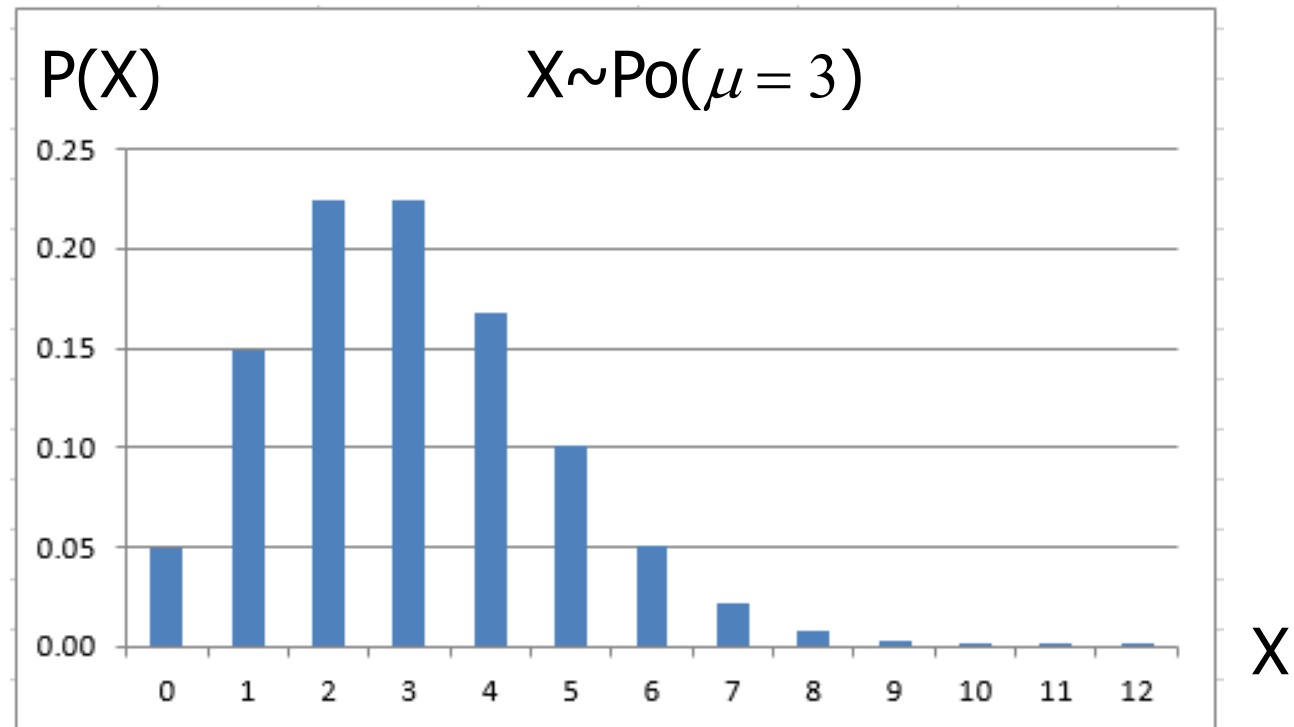
- The expected value and the variance of X are both equal to μ .

Prob. Distribution of Discrete r.v.

Poisson Distribution – $X \sim \text{Pois}(\mu)$

$$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$$

Example:



Prob. Distribution of Discrete r.v.

Eg:

A salesman sells on the average 3 iphones per day. The number of iphones sold per day is a Poisson r.v.

- (i) Calculate the probability that in a given day he will sell some iphones.
- (ii) Given that there are 8 working hours per day, what is the probability that in a given hour he will sell one iphone?

Prob. Distribution of Discrete r.v.

Soln:

(i) $P(\text{some iphones}) = 1 - P(\text{no iphone})$

$$= 1 - \frac{e^{-3} 3^0}{0!} = 0.95$$

(ii) Avg no. of iphones sold/hour = $3/8 = 0.375$

On a given hour, $P(X=1) = \frac{e^{-0.375} 0.375^1}{1!}$

Wooclap Ex: A shop selling a certain product of which the weekly demand is a Poisson variable with mean 3. On 1st day of each month, the stocks are replenished. Obtain the minimum number of the product in stock on 1st day of a month so that the shop is at least 50% sure of being able to meet the demands for the month.

Prob. Distribution of Discrete r.v.

Poisson Distribution – Approx to Binomial Dist

For Binomial Distribution $X \sim B(n, p)$ with large n :
i.e. $n \rightarrow \infty$, $p \rightarrow 0$ and $np \rightarrow$ a constant μ

Then it can be shown that :

$$\frac{n!}{x! (n-x)!} p^x (1-p)^{n-x} \rightarrow \frac{e^{-\mu} \mu^x}{x!}$$

$\therefore X \sim B(n, p) \approx \text{Pois}(\mu)$, where $\mu = np$, we have:

$$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$$

Prob. Distribution of Discrete r.v.

Poisson Distribution – Approx to Binomial Dist

Conditions for which the probability of $X \sim B(n, p)$ can be approximated by the probability $X \sim \text{Pois}(\mu)$:

- large n ($n \geq 100$)
- small p ($p \leq 0.01$)
- constant $\mu = np$ ($\mu \leq 20$)

Note that when $p \rightarrow 0$, then $(1 - p) \rightarrow 1$.

$\therefore$ the expected value and the variance of $X \sim B(n, p)$ are approximately equal.

Intuitive Explanation:

To obtain the mean and variance of the Poisson distribution, consider the binomial distribution under the following conditions:

$n \rightarrow \infty$, $p \rightarrow 0$ and $np \rightarrow \text{a constant } \mu$

The mean and variance of the binomial distribution are given by:

$$E[X] = np$$

$$\text{Var}[X] = npq$$

Since $q = 1 - p$, and as $p \rightarrow 0$, $q \rightarrow 1$ we have:

$$E[X] = \mu \text{ and } \text{Var}[X] = npq = \mu$$

Example: Suppose in a production line, 1 in 200 LED produced is defective. A random sample of 1000 LEDs are selected. What is the probability that at most 2 of them are defective?

Let X = no. of defective LED

p = Prob that a LED is defective = $1/200 = 0.005$

$np = 5$

$n > 100$, $p < 0.01$ and $np < 20$

use Poisson Dist to approx. Binomial Dist

$$\begin{aligned} P(X \leq 2) &= P(X=0) + P(X=1) + P(X=2) \\ &= e^{-5} + 5e^{-5} + \frac{5^2 e^{-5}}{2!} = 0.125 \end{aligned}$$

Prob. Distribution of Discrete r.v.

Special Probability Distributions

- Binomial Distribution
- Poisson Distribution
- Geometric Distribution

Prob. Distribution of Discrete r.v.

Geometric Distribution – Notation: $X \sim G(p)$

- Consider a sequence of independent Bernoulli trials, where each trial is an “experiment” with exactly 2 possible outcomes, “success” and “failure”.
- Assume that the probability of success (S) is the same for all trials, $P(S) = p$, and the probability of failure $P(F) = 1 - p$.
- Each experiment is performed consecutively until the first success is obtained.
- Let X = no. of trials. Hence there are $X - 1$ failures before the 1st success is obtained.

Prob. Distribution of Discrete r.v.

Hence X is a random variable with a Geometric distribution $X \sim G(p)$:

$$P(X = x) = (1 - p)^{x-1} p$$

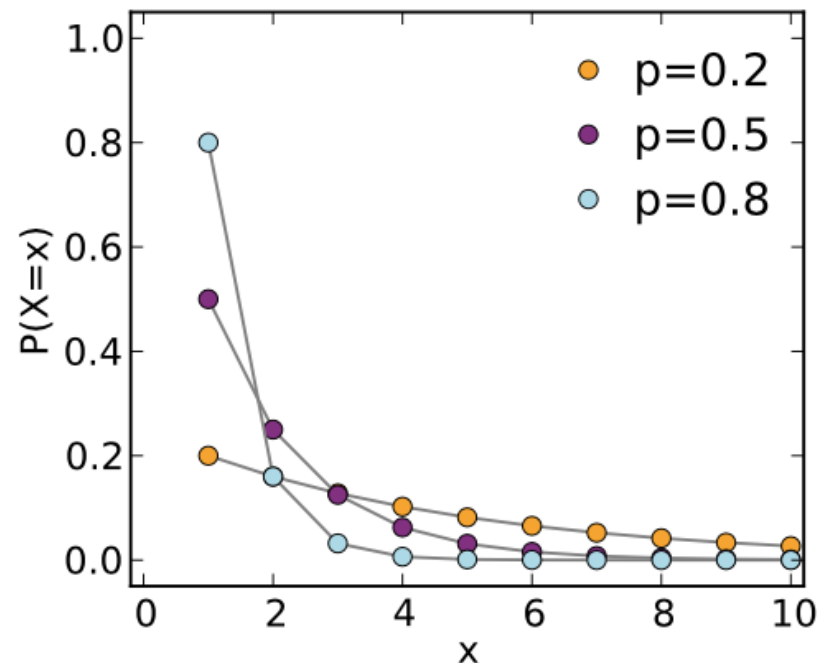
Expected value:

$$E[X] = \frac{1}{p}$$

Variance:

$$\text{Var}[X] = \frac{1 - p}{p^2}$$

Examples:



Example:

In a data transmission, the probability that a data frame is received in error is 0.1. If a data frame is received in error, a retransmission will be requested.

- a) What is probability that a data frame will be successfully received in no more than 3 transmissions?
- b) Determine the average number of transmissions per frame.

Let X = # of data frames transmitted to get the 1st success

$$\therefore X \sim G(p = 0.9)$$

$$\begin{aligned} \text{a) } P(X \leq 3) &= P(X=1) + P(X=2) + P(X=3) \\ &= 0.9 + 0.1 * 0.9 + 0.1^2 * 0.9 = 0.999 \end{aligned}$$

$$\text{b) } E[X] = 1/p = 1.111$$

Prob. Distribution of Discrete r.v.

Special Probability Distributions

- Binomial Distribution
- Poisson Distribution
- Geometric Distribution

Distribution	Probability	Mean	Variance
$X \sim B(n, p)$	$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$	np	$np(1 - p)$
$X \sim \text{Pois}(\mu)$	$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$	μ	μ
$X \sim G(p)$	$P(X = x) = (1 - p)^{x-1} p$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$